

CD8+ T Cells and NK Cells: Parallel and Complementary Soldiers of Immunotherapy.

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CD8+ T cells and NK cells are both cytotoxic effector cells of the immune system, but the recognition, specificity, sensitivity, and memory mechanisms are drastically different. While many of these topics have been extensively studied in CD8+ T cells, very little is known about NK cells. Current cancer immunotherapies mainly focus on CD8+ T cells, but have many issues of toxicity and efficacy. Given the heterogeneous nature of cancer, personalized cancer immunotherapy that integrates the power of both CD8+ T cells in adaptive immunity and NK cells in innate immunity might be the future direction, along with precision targeting and effective delivery of tumor-specific, memory CD8+ T cells and NK cells.

Differential expression of granzymes A and B in human cytotoxic lymphocyte subsets and T regulatory cells.

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Cytotoxic T lymphocytes (CTLs) and natural killer (NK) cells use the perforin/granzyme pathway as a major mechanism to kill pathogen-containing cells and tumor cells.(1,2) Dysregulation of this pathway results in several human diseases, such as hemophagocytic lymphohistiocytosis. Here we characterize the single-cell expression pattern of granzymes A and B in human lymphocytes using a flow cytometry-based assay. We demonstrate that most circulating CD56(+)8(-) NK cells, and approximately half of circulating CD8(+) T lymphocytes, coexpressed both granzymes A and B. In contrast, few circulating CD4(+) T lymphocytes expressed granzymes A or B. Activation of CD8(+) T lymphocytes with concanavalin A (ConA)/interleukin-2 (IL-2), and activation of CD4(+) T lymphocytes with antibodies to CD3/CD28 or CD3/CD46 (to generate T regulatory [Tr1] cells), induced substantial expression of granzyme B, but not granzyme A. Naive CD4(+)CD45RA(+) cells stimulated with antibodies to CD3/CD46 strongly expressed granzyme B, while CD3/CD28 stimulation was ineffective. Finally, we show that granzyme B-expressing CD4(+) Tr1 cells are capable of killing target cells in a perforin-dependent, but major histocompatibility complex (MHC)/T-cell receptor (TCR)-independent, manner. Our results demonstrate discordant expression of granzymes A and B in human lymphocyte subsets and T regulatory cells, which suggests that different granzymes may play unique roles in immune system responses and regulation.

Differential expression of human granzymes A, B, and K in natural killer cells and during CD8+ T cell differentiation in peripheral blood.

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NK cells and cytotoxic T lymphocytes can induce apoptosis in virus-infected and transformed target cells via the granule exocytosis pathway. The key components of the cytolytic granules are perforin and several serine esterases, termed granzymes. While the cellular distribution of human granzymes A (GrA) and B (GrB) has been well characterized much less is known about the expression pattern of human granzyme K (GrK). In this study GrA, GrB, and GrK expression was analyzed in human peripheral blood lymphocytes using flow cytometry. There was a distinct population of GrK expressing CD8+ T cells with a CD27+/CD28+/CCR5high/CCR7-/perforin-/low/IFN-gamma+ memory-like phenotype, while all CD56bright

NK cells were also positive for GrK. In addition, GrK was also expressed in subpopulations of CD56+ T cells, CD4+ T cells, and TCRgammadelta+ T cells. In contrast, GrB was primarily expressed in CD56dim NK cells and differentiated memory CD8+ T cells with the CD27-/low/CD28-/low/CCR5-/low/CCR7-/CD11b+/perforinhigh phenotype. Only few CD8+ T cells expressed both GrB and GrK. GrA was found to be co-expressed in all GrB- and GrK-expressing T cells. Our findings suggest that granzyme expression during the differentiation process of memory CD8+ T cells might be as follows: GrA+/GrB-/GrK+ --> GrA+/GrB+/GrK+ --> GrA+/GrB+/GrK-.

Deep immune profiling by mass cytometry links human T and NK cell differentiation and cytotoxic molecule expression patterns.

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The elimination of infected or tumor cells by direct lysis is a key T and NK cell effector function. T and NK cells can kill target cells by coordinated secretion of cytotoxic granules containing one or both pore-forming proteins, perforin and granulysin and combinations of granzyme (Gzm) family effector proteases (in humans: Gzm A, B, K, M and H). Understanding the pattern of expression of cytotoxic molecules and the relationship to different states of T and NK cells may have direct relevance for immune responses in autoimmunity, infectious disease and cancer. Approaches capable of simultaneously evaluating expression of multiple cytotoxic molecules with detailed information on T and NK differentiation state, however, remain limited. Here, we established a high dimensional mass cytometry approach to comprehensively interrogate single cell proteomic expression of cytotoxic programs and lymphocyte differentiation. This assay identified a coordinated expression pattern of cytotoxic molecules linked to CD8 T cell differentiation stages. Coordinated high expression of perforin, granulysin, Gzm A, Gzm B and Gzm M was associated with markers of late effector memory differentiation and expression of chemokine receptor CX3CR1. However, classical gating and dimensionality reduction approaches also identified other discordant patterns of cytotoxic molecule expression in CD8 T cells, including reduced perforin, but high Gzm A, Gzm K and Gzm M expression. When applied to non-CD8 T cells, this assay identified different patterns of cytotoxic molecule co-expression by CD56hi versus CD56dim defined NK cell developmental stages; in CD4 T cells, low expression of cytotoxic molecules was found mainly in TH1 phenotype cells, but not in Tregs or T follicular helper cells (TFH). Thus, this comprehensive, single cell, proteomic assessment of cytotoxic protein co-expression patterns demonstrates specialized cytotoxic programs in T cells and NK cells linked to their differentiation stages. Such comprehensive cytotoxic profiling may identify distinct patterns of cytotoxic potential relevant for specific infections, autoimmunity or tumor settings.

Absence of cytotoxic molecules in CD8- and/or CD56-positive adult T-cell leukaemia/lymphoma.

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Adult T-cell leukaemia/lymphoma (ATLL) cells usually exhibit a CD4+ (helper/inducer) phenotype (CD4+/8-/56-), and only a minority of tumours express the CD8 (cytotoxic/suppressor) or CD56 (natural killer [NK]-associated) antigens. TIA-1 is a cytotoxic granule-associated protein expressed in NK cells and cytotoxic T lymphocytes (CTLs). Granzyme B, perforin and Fas ligand (FasL) are also expressed in

activated CTLs and NK cells. To clarify the cytotoxic potential of ATLL cells, immunohistochemistry was performed in CD8+ and/or CD56+ ATLL cells, using anti-TIA-1, anti-granzyme B, anti-perforin and anti-FasL antibodies. We studied nine cases of CD8+ and/or CD56+ ATLL, all of which exhibited monoclonal integration of human T-cell leukaemia virus type 1 (HTLV-1) proviral DNA. Four cases exhibited a CD8+/CD56- phenotype, four others had a CD8-/CD56+ phenotype, and one was CD8+/CD56+. All but one case also expressed the surface antigens CD3, TCR alpha beta, and CD4. Expression of granzyme B and TIA-1 were demonstrated in three and two cases, respectively, but none expressed perforin or FasL. In the control study, 10 cases with typical CD3+/4+/8-/56- ATLL demonstrated no expression of those cytotoxic-associated proteins. Our findings suggest that CD8 and/or CD56 positivity probably confer(s) no cytotoxic function on ATLL cells, and it is possible that CD8 and CD56 may be simply aberrant surface markers in ATLL.

Granzyme B and perforin are important for regulatory T cell-mediated suppression of tumor clearance.

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Granzyme B is important for the ability of NK cells and CD8(+) T cells to kill their targets. However, we showed here that granzyme B-deficient mice clear both allogeneic and syngeneic tumor cell lines more efficiently than do wild-type (WT) mice. To determine whether regulatory T (Treg) cells utilize granzyme B to suppress immune responses against these tumors, we examined the expression and function of granzyme B in Treg cells. Granzyme B was not expressed in naive Treg cells but was highly expressed in 5%-30% of CD4(+)Foxp3(+) Treg cells in the tumor environment. Adoptive transfer of WT Treg cells, but not granzyme B- or perforin-deficient Treg cells, into granzyme B-deficient mice partially restored susceptibility to tumor growth; Treg cells derived from the tumor environment could induce NK and CD8(+) T cell death in a granzyme B- and perforin-dependent fashion. Granzyme B and perforin are therefore relevant for Treg cell-mediated suppression of tumor clearance in vivo.

Sinonasal T-cell expression of cytotoxic mediators granzyme B and perforin is reduced in patients with chronic rhinosinusitis.

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CD8+ T cells and natural killer (NK) cells are cytotoxic cells that use granzyme B (GrB) and perforin. Defective cytotoxic function is known to play a role in dysregulated immune response as seen in chronic sinusitis, also referred to as chronic rhinosinusitis (CRS). However, to our knowledge, in the United States, neither GrB or perforin expression has been reported in patients with CRS. The aim of this study was to investigate sinonasal cytotoxic cells, their mediators, and cell-specific distribution of these mediators in patients with CRS with nasal polyp (CRSwNP) and in patients with CRS without nasal polyp (CRSsNP). Blood and sinus tissue samples were taken from patients with CRSsNP (n = 8) and CRSwNP (n = 8) at the time of surgery. Control subjects (n = 8) underwent surgery for cerebrospinal fluid leak repair or to remove non-hormone-secreting pituitary tumors. The cells were analyzed via flow cytometry by using CD8 expression to identify cytotoxic T cells and CD56 expression to identify NK cells. Intracellular GrB and perforin expression were analyzed with flow cytometry. We observed no significant differences in plasma or peripheral blood immune cell numbers or specific levels of GrB or perforin among the groups.

In the sinonasal mucosa of the patients with CRSsNP and the patients with CRSwNP, there was a significant decrease in GrB and perforin levels ($p < 0.05$) despite similar or increased numbers of cytotoxic cells when compared with the controls. The overall decrease in GrB and perforin in the sinonasal mucosa of the patients with CRSsNP and the patients with CRSwNP was due to decreased T cell production. There was no difference in total NK cell count or expression of perforin or GrB among all the groups. Total levels of sinonasal GrB and perforin were decreased in the sinonasal mucosa of both the patients with CRSwNP and the patients with CRSsNP compared with the controls, whereas sinonasal CD8+ T cells, (but not NK cells,), intracellular stores of GrB and perforin were reduced in the patients with CRSwNP compared with the controls.

Decreased levels of perforin-positive lymphocytes are associated with posttraumatic complications in patients with major trauma.

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Posttraumatic immune disorder can cause complications including systemic inflammatory response syndrome (SIRS) and multiple-organ dysfunction syndrome (MODS). Cytotoxic granules containing perforin and granzyme-B (GrB) are released by cytotoxic CD8(+) T lymphocytes, NK and $\gamma\delta$ T cells after major trauma. This prospective clinical study was designed to analyze the association between these immune components and complications after major trauma. We retrospectively studied 48 patients aged between 16 and 65 years admitted within 90min of major trauma (Injury Severity Score >16) and surviving beyond 7 days, and 20 healthy controls. Blood samples were drawn on admission and after 1, 3 and 7 days. CD8(+) T, NK and $\gamma\delta$ T cell counts in peripheral blood and the levels of perforin and GrB in these cells were analyzed by flow cytometry. Clinical aspects of MODS and SIRS were recorded daily. CD8(+) T cell counts were not significantly different in patients with SIRS or uncomplicated group, but were depressed in the MODS group after trauma. However, NK cell counts in patients with MODS were significantly depressed only at day 7 after injury, and $\gamma\delta$ T cell counts were significantly depressed after trauma. Perforin levels in CD8(+) T, NK and $\gamma\delta$ T cells in patients with MODS were depressed after trauma. GrB levels in NK, CD8(+) T and $\gamma\delta$ T cells in patients with MODS were significantly depressed at 3 and 7 days post trauma. Posttraumatic MODS is associated with early, sustained, and severe depression of lymphocytes.

Comparison of primary human cytotoxic T-cell and natural killer cell responses reveal similar molecular requirements for lytic granule exocytosis but differences in cytokine production.

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Cytotoxic lymphocytes, encompassing cytotoxic T lymphocytes (CTLs) and natural killer (NK) cells, kill pathogen-infected, neoplastic, or certain hematopoietic cells through the release of perforin-containing lytic granules. In the present study, we first performed probability-state modeling of differentiation and lytic granule markers on CD8(+) T cells to enable the comparison of bona fide CTLs with NK cells. Analysis identified CD57(bright) expression as a reliable phenotype of granule marker-containing CTLs. We then compared CD3(+)CD8(+)CD57(bright) CTLs with NK cells. Healthy adult peripheral blood CD3(+)CD8(+)CD57(bright) CTLs expressed more granzyme B but less perforin than CD3(-)CD56(dim) NK

cells. On stimulation, such CTLs degranulated more readily than other T-cell subsets, but had a propensity to degranulate that was similar to NK cells. Remarkably, the CTLs produced cytokines more rapidly and with greater frequency than NK cells. In patients with biallelic mutations in UNC13D, STX11, or STXBP2 associated with familial hemophagocytic lymphohistiocytosis, CTL and NK cell degranulation were similarly impaired. Therefore, cytotoxic lymphocyte subsets have similar requirements for Munc13-4, syntaxin-11, and Munc18-2 in lytic granule exocytosis. The present results provide a detailed comparison of human CD3(+)CD8(+)CD57(bright) CTLs and NK cells and suggest that analysis of CD57(bright) CTL function may prove useful in the diagnosis of primary immunodeficiencies including familial hemophagocytic lymphohistiocytosis.

Expression of granzyme B in peripheral blood polymorphonuclear neutrophils (PMN), myeloid cell lines and in PMN derived from haematopoietic stem cells in vitro.

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Granzyme B and perforin are the major protagonists of cytotoxicity mediated by natural killer (NK) cells or cytotoxic T cells. More recent we described the presence of granzyme B and perforin in polymorphonuclear neutrophils (PMN), a finding in discrepancy with the credo that granzyme B and perforin expression is restricted to cytotoxic T cells and NK cells. In extension of our previous study, we now provide evidence that granzyme B is not only present in mature PMN, but also in the myeloid cell lines HL-60 and U937, in CD34+ stem cells, and in PMN derived from CD34+ cells in vitro. In agreement with the "targeting by time" hypothesis we found the bulk of granzyme B in association with primary granules, in addition to a minor membrane expression. Granzyme B, on one hand might, enhance the cytotoxic potential of PMN, on the other, it may provide PMN with additional means to degrade extracellular matrices.